CREDITLENS

Executive Summary — Credit Risk & Default Analysis

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GitHub: GitHub Repository
Tableau Public: CreditLens — Executive Risk Report
Streamlit App: Borrower Risk Assessment App

Technology: PostgreSQL · SQL · Python · Pandas · scikit-learn · Tableau Public · Streamlit

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Bottom Line

Across 32,581 loan records representing **USD 312.43 million in recorded loan amount**, 7,108 loans are classified as defaults, producing an overall Default Rate of **21.82%**. The model-based Expected Loss estimate is **USD 57.30 million**, calculated using Logistic Regression scores and an assumed 45% loss-given-default.

Business Context

The challenge

Portfolio-level averages can conceal material differences in default risk across credit grades, income groups and borrower affordability profiles. Risk teams need a clearer view of where defaults concentrate and which applications may warrant additional scrutiny.

The approach

CreditLens combines structured data validation, PostgreSQL risk analysis, Logistic Regression scoring and interactive Tableau and Streamlit reporting. The analysis examines observed portfolio outcomes and uses model scores to estimate illustrative Expected Loss.

Key Findings

1. **Credit grade is a strong risk differentiator.** Default Rate rises from 9.96% for Grade A to 59.05% for Grade D and 98.44% for Grade G.
2. **Income cohorts differ materially.** The under-25K income group records a 54.44% Default Rate, compared with 10.02% for the 200K+ group.
3. **Higher loan burden corresponds with higher observed default.** The 40%+ loan-burden cohort has a 38.87% Default Rate, compared with 11.07% for the under-20% cohort.
4. **Default rates vary by loan purpose.** Debt consolidation records 28.59%, Medical 26.70% and Home Improvement 26.10%.

5. **The model shows promising holdout discrimination.** Logistic Regression achieved a recorded ROC-AUC of 0.8784.

Recommendations

1. **Strengthen affordability checks:** Refer 40%+ loan-burden cases and high-burden/weak-grade combinations for enhanced review; assess whether lower initial exposure is appropriate.
2. **Combine risk signals:** Consider model scores alongside credit grade, affordability and existing underwriting information when evaluating pricing, exposure limits or manual review.
3. **Test structured review tiers:** Evaluate and validate score bands below 20%, 20–45% and above 45% as case-study review categories.
4. **Investigate cohort differences:** Use income and loan-purpose patterns to guide further analysis, not as standalone eligibility or exclusion rules.
5. **Monitor and recalibrate:** Track manual overrides and subsequent outcomes, evaluate performance across borrower segments, and update controls only when sufficient evidence supports a change.

Limitations and Data Quality

- The recorded loan amount is not verified disbursed capital. The loan-burden field used for cohort analysis is not automatically equivalent to conventional total-debt DTI.
- The 45% LGD is an assumption. The USD 57.30 million Expected Loss estimate is not realized loss or proven savings.
- ROC-AUC measures discrimination, not probability calibration. The model is not validated for real lending decisions.
- Small cohorts, including Grade G, require cautious interpretation. Observed differences do not establish causation.

Conclusion

CreditLens demonstrates an end-to-end analytical workflow connecting data quality, SQL, risk segmentation, model evaluation and business reporting. Its practical contribution is a clearer view of default concentration and a set of underwriting controls that can be tested and monitored, not a claim of realized loss reduction or production-ready lending decisions.